

Blind dereverberation exploiting Cross-Relation of Monoral System on the TF-representation

```
clear
```

settings

```
% DGT tool
F = DGTtool("windowLength", 512, "windowShift", 128);

% audio data loading
%path = "input/4077_13754_2sec_11025";
path = "input/Goldberg_clav";
fpaths = dir(path);
ss = cell(2,1);
for i=1:2
    audiofilename = path + "/" + fpaths(i+2).name;
    [temp, fs] = audioread(audiofilename);
    noise = randn(size( temp(:,1) ));
    noise = 0.1 * noise * norm(temp(:,1)) / norm(noise);
    ss{i} = temp(:,1) + noise;
end
snr(temp(:,1), noise)
```

```
ans =
20
```

```
lambda = 1e-10;
s = [ss{1}; ss{2}];
num_mic = 1;

% ir data loading
% ir_data_path = "BFTLSPACELIBRARY_ChamberAL_11025.wav"; % RT60 = 1.89
%ir_data_path = "../data/wavIR/IR_sweep_15s_45Hzto22kHz_FS16kHz.v370.wav" %
%RT60 = 0.45
ir_data_path = "../data/wavIR/01.0s Villa Bathroom-OST.wav" % RT60 = 1.0
```

```
ir_data_path =
"../data/wavIR/01.0s Villa Bathroom-OST.wav"
```

```
[ir, ~] = audioread(ir_data_path);
%[ir, fs] = audioread("logic03sWoodenBooth8000.wav");
%[ir, fs] = audioread("../data/BF TL SPACE LIBRARY/Drumbrella/Drumbrella 5'.R.wav");
```

Malloc

```
rt = zeros(1,1); % reverb time in sec
obs = cell(1,2); % obs
X = cell(1, 2); % STFTs of obs. signal
S = cell(1, 2); % STFTs of dry signal
```

```

N = zeros(1, 2); % length of obserbation
Lir = zeros(1, 1); % ir length = d + 1
l = zeros(1, 2); % length of deconvoluted signal

```

resample and shortening

```

% calculate the reververation time (RT60)
rt = reverb_time(ir, fs);
ir = ir(1:floor(fs*rt)); ir = ir(:) / norm(ir);
IR = F(ir);
Lir = size(IR, 2);

for j = 1:2
    obs{j} = conv(ss{j}, ir);
    X{j} = F(obs{j});
    N(j) = size(X{j}, 2);
    S{j} = F(ss{j});
end
obsCat = [obs{1}; obs{2}];
D = Lir - 1;
padding = -2;
D = D + padding;
Lir = Lir + padding;

```

main

```

%% proposed
% memory allocation
Fq = size(X{1}, 1);
for i = 1:2, l(i) = N(1, i) - Lir(1) + 1; end
RET1 = zeros(Fq, l(1)); RET2 = zeros(Fq, l(2));

% frequency-wise processing
tic
parfor f = 1:Fq
    temp = X{1, 1}; X1f = temp(f, :)';
    temp = X{1, 2}; X2f = temp(f, :)';
    Xf = [X1f; X2f];
    XfNorm = norm(Xf); XfPhase = Xf ./ abs(Xf);

    % Null-space calculation via (not structured) low-rank approximation
    Sylvester = [convmtx(X2f, l(1)) convmtx(-X1f, l(2))];
    [~, ~, V] = svd(Sylvester, "econ");
    Vf = V(:,end);
    Vf = XfNorm * Vf / norm(Vf); % scale recovery
    v1 = Vf(1:l(1)); v2 = Vf(l(1)+1:end);
    RET1(f,:) = v1'; RET2(f,:) = v2';
end
% inverse Descrete Gabor Transoformation

```

```
ret = cell(1);
ret{1} = F.pinv(RET1); ret{2} = F.pinv(RET2);
```

index computed (OLA)

```
t_CR = toc;
```

```
%% lifting
ret_lift = cell(2,1);
RET_lift = cell(2,1);
tic
for i = 1:2
    speech_data_path = (path + "/" + fpaths(i+2).name);
    RET_lift{i} = dereverb(speech_data_path, ir_data_path);
    ret_lift{i} = F.pinv(RET_lift{i});
end
```

警告: コロン演算子をインデックスとして使う場合、整数オペランドが必要です。

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```
t_lift = toc;

%% WPE
ret_wpe = cell(2,1);
tic
for i = 1:2
    % Perform WPE (Weighted Prediction Error) processing
    ret_wpe{i} = WPE1c_wrapper(obs{i}, fs);
end
```

```
[Data information]
Audio size          : 5.58 sec
Sampling frequency  : 11 kHz
Number of microphones : 1
```

```
[Block information]
Number of blocks    : 1
Block size          : 190.05 sec
Shift size          : 189.75 sec
```

```
Processing block 1/1...
  1st iteration
  2nd iteration
  3rd iteration
```

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```
t_wpe = toc;
```

Results

```
for i=1:2
    len_ss = length(ss{i});
    len_ret = length(ret{i});
    if len_ss > len_ret
        ss{i} = ss{i}(1:len_ret);
        len_ss = len_ret;
    end
    [SDR,SIR,SAR,perm] = bss_eval_sources(obs{i}(1:len_ss)', ss{i}');
    fprintf("obs: SDR %2.3f, SIR %2.3f, SAR %2.3f\n", SDR, SIR, SAR)

    [SDR,SIR,SAR,perm] = bss_eval_sources(ret{i}(1:len_ss)', ss{i}');
    fprintf("ret: SDR %2.3f, SIR %2.3f, SAR %2.3f\n", SDR, SIR, SAR)

    [SDR,SIR,SAR,perm] = bss_eval_sources(ret_lift{i}(1:len_ss)', ss{i}');
    fprintf("ret(lift): SDR %2.3f, SIR %2.3f, SAR %2.3f\n", SDR, SIR, SAR)

    [SDR,SIR,SAR,perm] = bss_eval_sources(ret_wpe{i}(1:len_ss)', ss{i}');
    fprintf("ret(wpe): SDR %2.3f, SIR %2.3f, SAR %2.3f\n", SDR, SIR, SAR)

    %audiowrite("output/" + sprintf(audiofilename + "_wet.wav", i), obs{i}, fs)
    %audiowrite("output/" + sprintf(audiofilename + "_lra_ORACLE_pad=%d.wav", i,
padding), ret{i}, fs)
    %audiowrite("output/" + sprintf(audiofilename + "_lra_ORACLE(w)_pad=%d.wav", i,
padding), ret_w{i}, fs)
    % audiowrite("output/slra.wav", ret_slra, fs)
end
```

```
obs: SDR 1.338, SIR Inf, SAR 1.338
ret: SDR 6.356, SIR Inf, SAR 6.356
ret(lift): SDR 2.568, SIR Inf, SAR 2.568
ret(wpe): SDR -0.008, SIR Inf, SAR -0.008
obs: SDR 0.687, SIR Inf, SAR 0.687
ret: SDR 5.169, SIR Inf, SAR 5.169
ret(lift): SDR 1.077, SIR Inf, SAR 1.077
ret(wpe): SDR -2.098, SIR Inf, SAR -2.098
```

```
fprintf("----Ex time----\n")
```

```
----Ex time----\n
```

```
fprintf("time: yohena : %f\n", sum(t_lift))
```

```
time: yohena      : 60.528283¥n
```

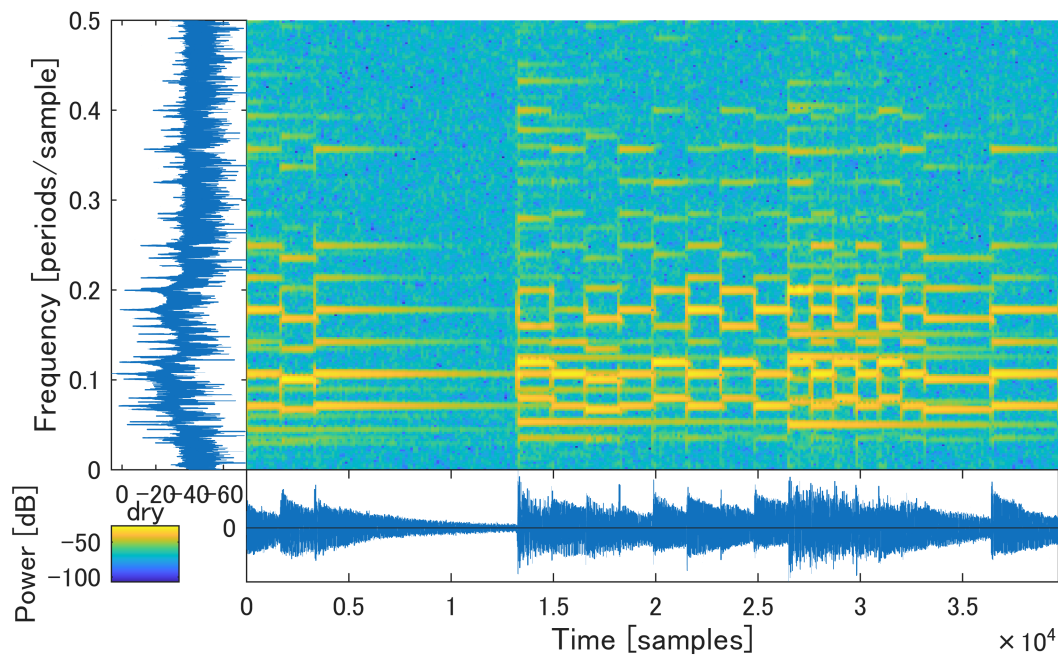
```
fprintf("time: monoral WPE : %f¥n", t_wpe)
```

```
time: monoral WPE : 0.536807¥n
```

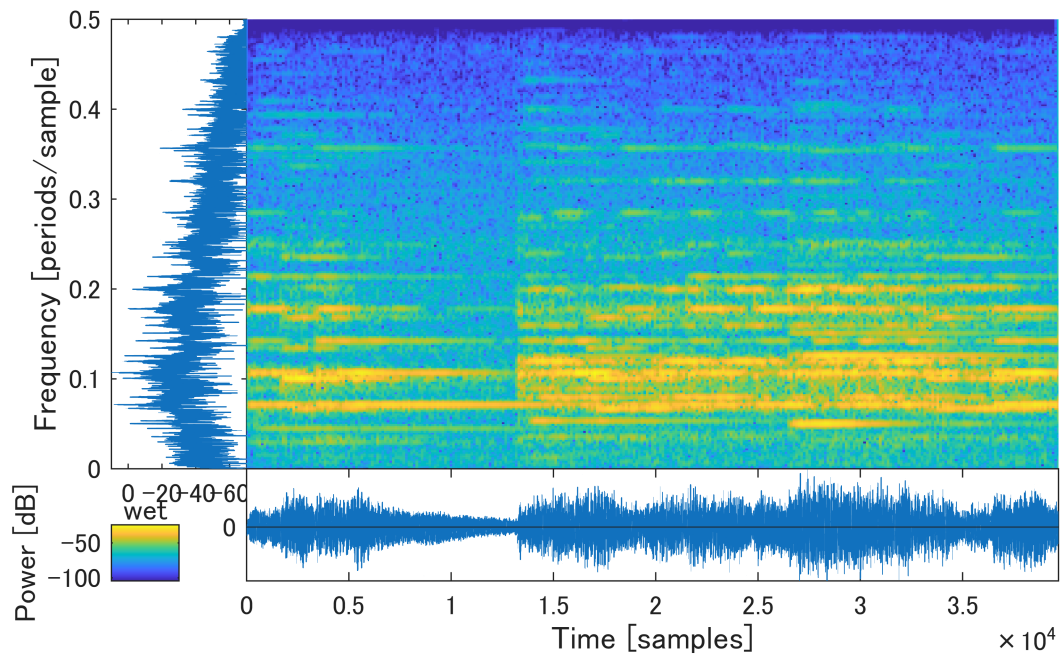
```
fprintf("time: monoral CR : %f¥n", t_CR)
```

```
time: monoral CR : 7.002687¥n
```

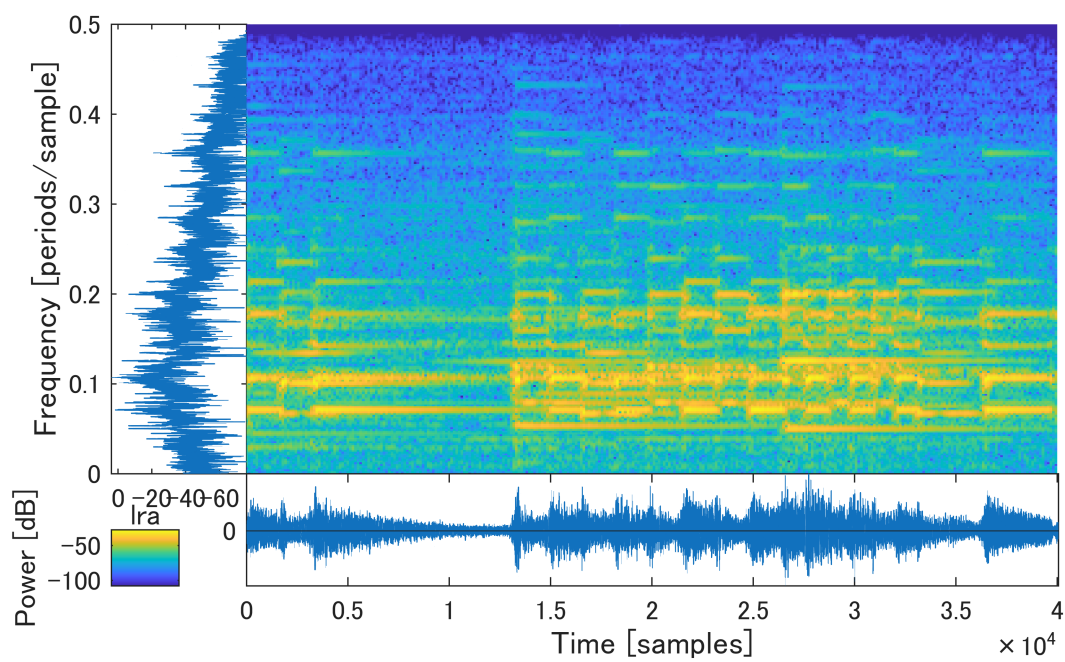
```
F.plot(ss{1}); title("dry");
```



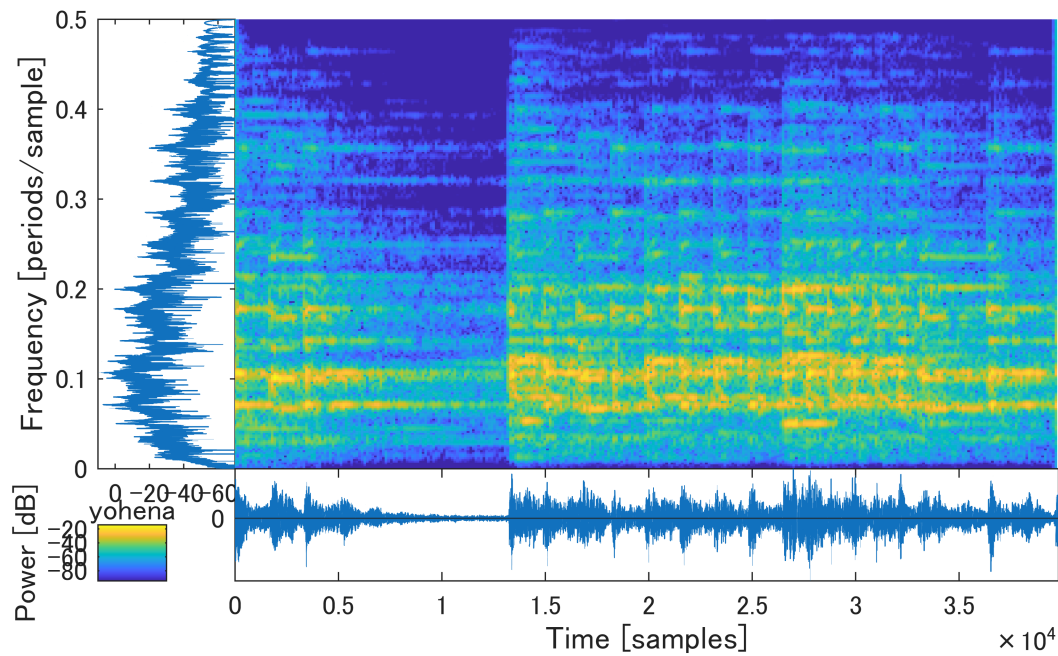
```
%saveas(gcf, "result/" + sprintf(audiofilename,1) + "_dry.png")  
F.plot(obs{1}(1:len_ss)); title("wet");
```



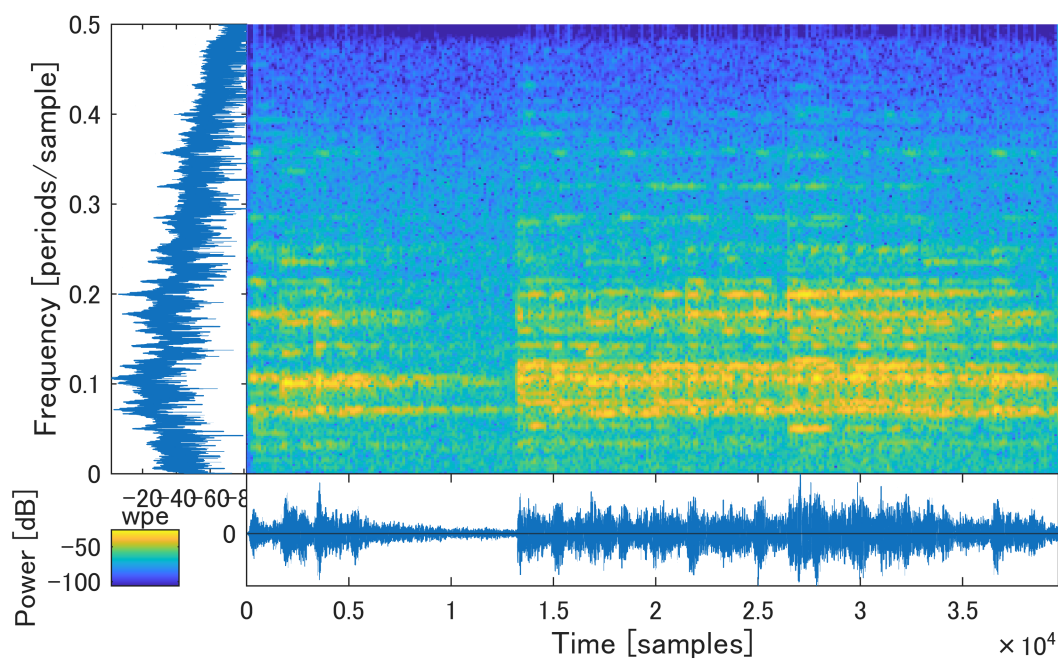
```
%saveas(gcf, "result/" + sprintf(audiofilename,1) + "_wet.png")
F.plot(ret{1}); title("lra");
```



```
%saveas(gcf, "result/" + sprintf(audiofilename + "_ORACLE_ret_pad=%d.png",1,
padding) )
F.plot(ret_lift{1}); title("yohena");
```

```
%saveas(gcf, "result/" + sprintf(audiofilename + "_ORACLE(w)_ret.png",1, padding) )
F.plot(ret_wpe{1}(1:len_ss)); title("wpe");
```



- どの結果も時間のマイナス方向に信号が漏れ出ているのが気になる。
- weight をいれることで若干改善。ただ、きいてもわからない